Introduction to Artificial Intelligence

Module 1

Think About It...

What comes to mind when you hear the term Artificial Intelligence?

- Robots?
- ChatGPT?
- Self-driving cars?
- Science fiction?
- Smart assistants?



Activity

- Discuss with a classmate.

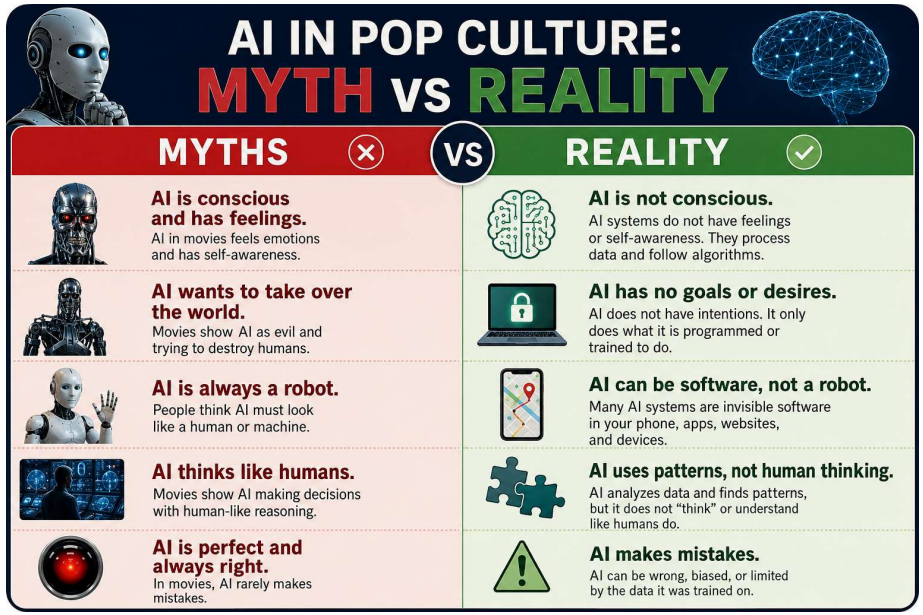
AI in Pop Culture: Myth vs Reality

Popular AI Myths:

- ❌ AI is conscious
- ❌ AI wants to take over the world
- ❌ AI is always a robot
- ❌ AI thinks like humans
- ❌ AI is always accurate

Reality:

- AI uses mathematics
- AI learns patterns
- AI follows algorithms
- AI makes mistakes



Why Should STEM Students Learn AI?

AI literacy is becoming as important as computer literacy.

AI is transforming:

- Engineering
- Healthcare
- Physics
- Biology
- Cybersecurity
- Environmental Science
- Manufacturing



Discussion

- How might AI affect your future career?

Learning Objectives

By the end of this module you should be able to:

- ✓ Define AI
- ✓ Explain AI history
- ✓ Distinguish AI, ML, and DL
- ✓ Describe Intelligent Agents
- ✓ Explain AGI
- ✓ Identify AI applications
- ✓ Discuss ethical concerns

What Is Artificial Intelligence?

Artificial Intelligence is:

The ability of computer systems to perform tasks that normally require human intelligence.

Examples:

- Learning
- Reasoning
- Problem solving
- Pattern recognition
- Language understanding
- Decision making

•Speed
•Large-scale data processing

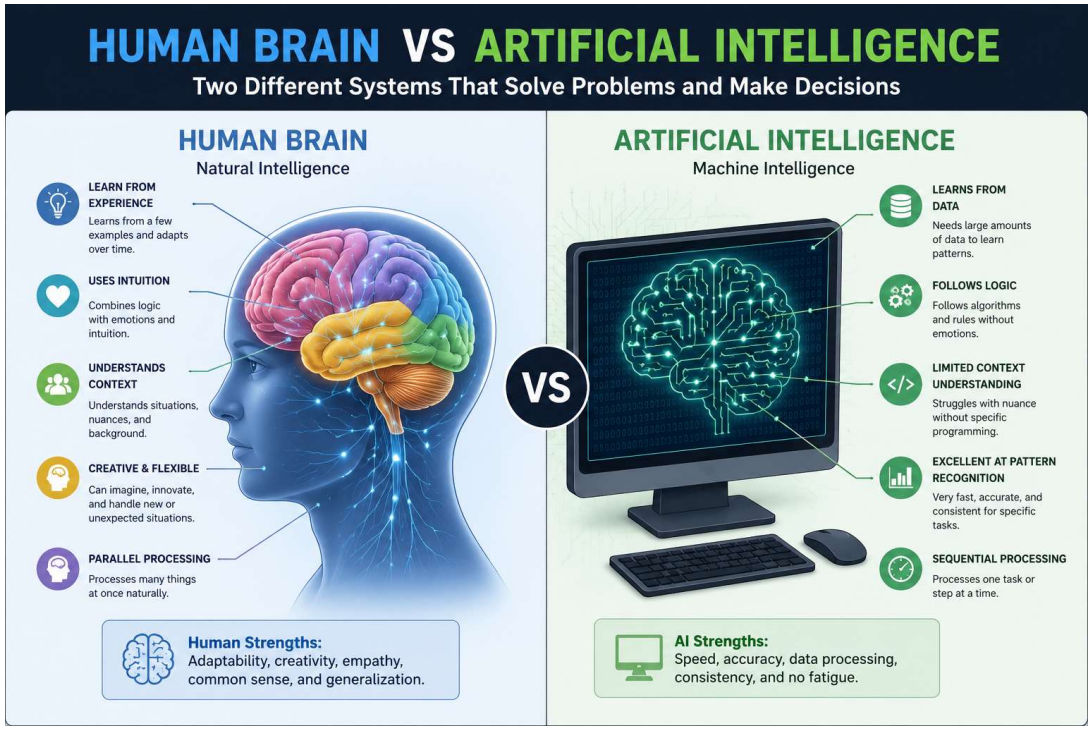
Human Intelligence vs Artificial Intelligence

Humans:

- Learn from experience
- Use intuition
- Understand context
- Creativity
- Common Sense
- Emotional Understanding

AI:

- Learns from data
- Recognizes patterns
- Makes predictions
- Pattern recognition
- Speed
- Large-scale data processing

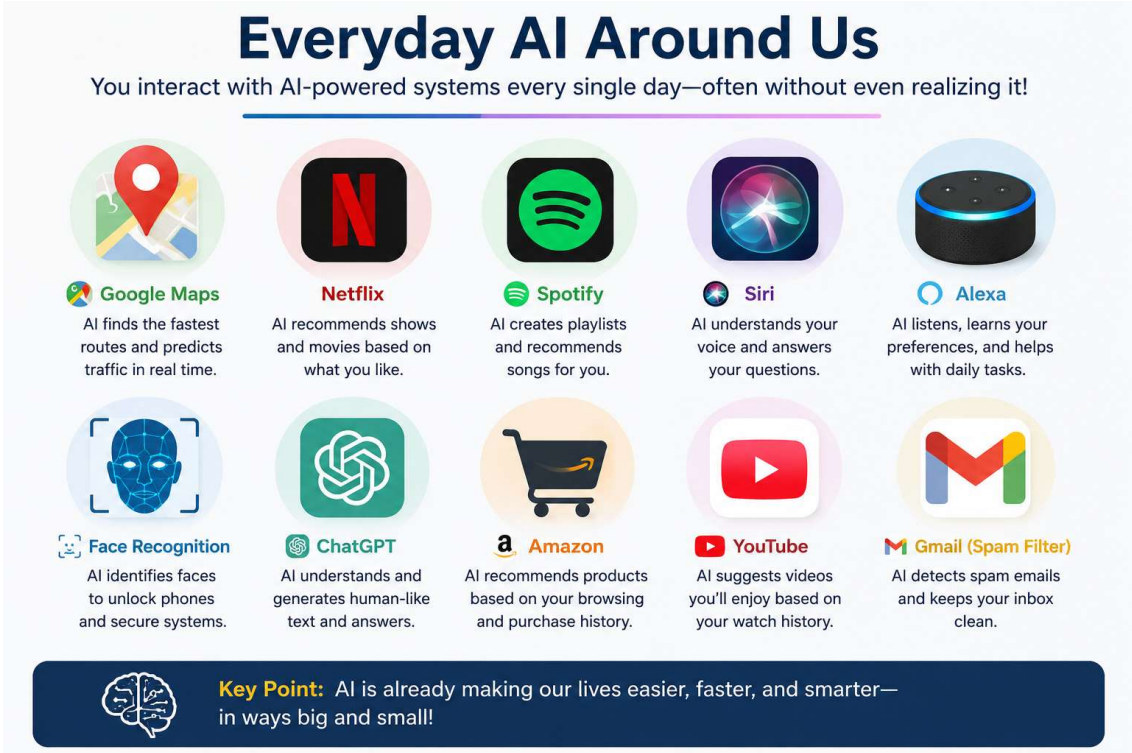


Everyday AI

How many AI systems have you used today?

Examples:

- Google Maps
- Netflix
- Spotify
- Siri
- Alexa
- Face ID
- ChatGPT



"Which AI system on this slide do you use most often?"

AI Is Already Everywhere

AI powers:

- Search engines
- Social media feeds
- Online shopping
- Healthcare systems
- Fraud detection
- Smart devices

Real-World Example

- Amazon recommendation engine

What AI Is NOT

AI is NOT:

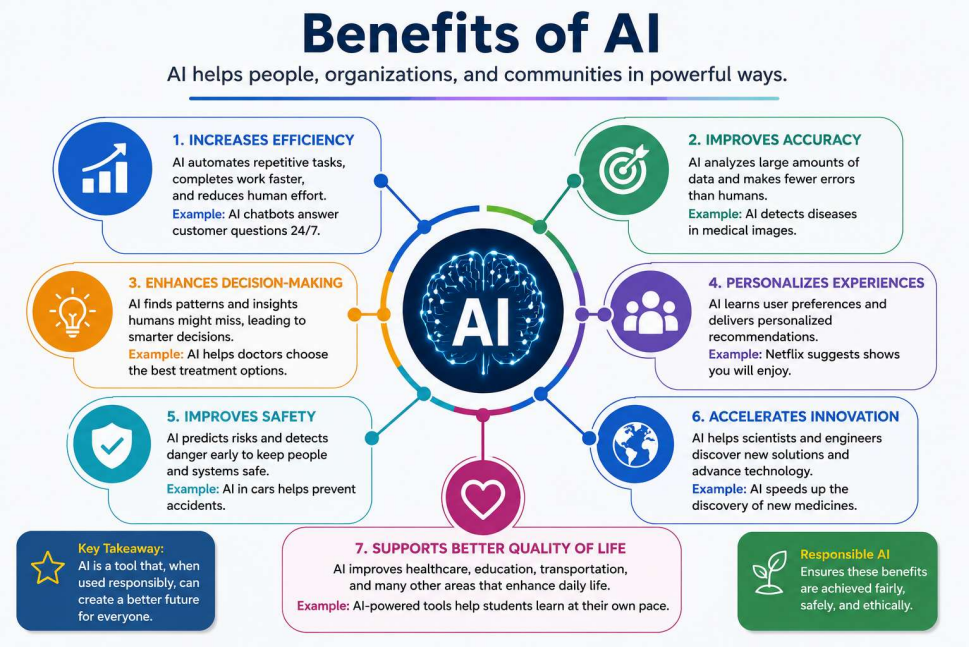
- ❌ Magic
- ❌ Conscious
- ❌ Human
- ❌ Perfect
- ❌ Always a robot

Key Idea: AI is software, data, mathematics, and computing power.

Benefits of AI

AI can:

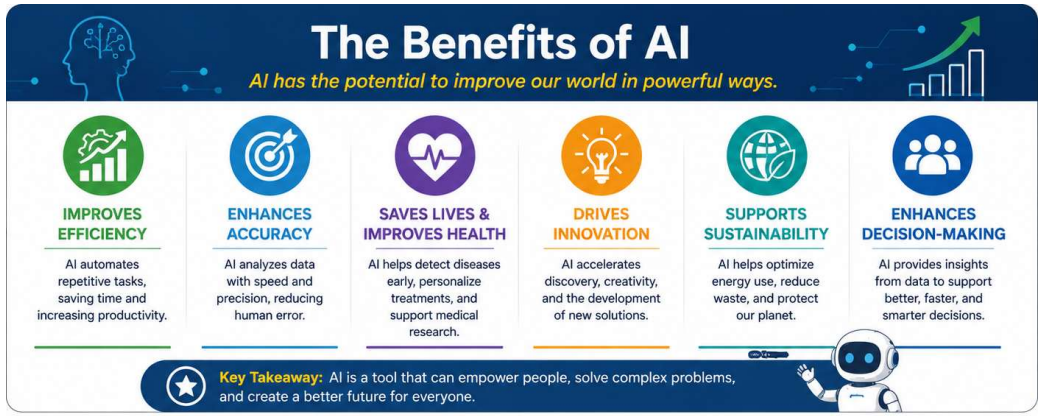
- Improve efficiency
- Increase productivity
- Detect diseases
- Improve safety
- Accelerate research
- Support decision-making
- Improves scientific discovery
- Helps solve complex problems

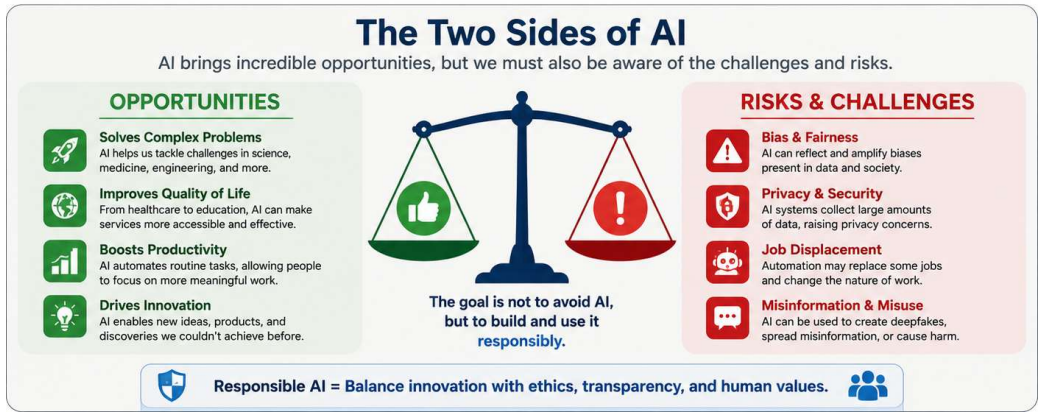


Challenges of AI

AI can also create:

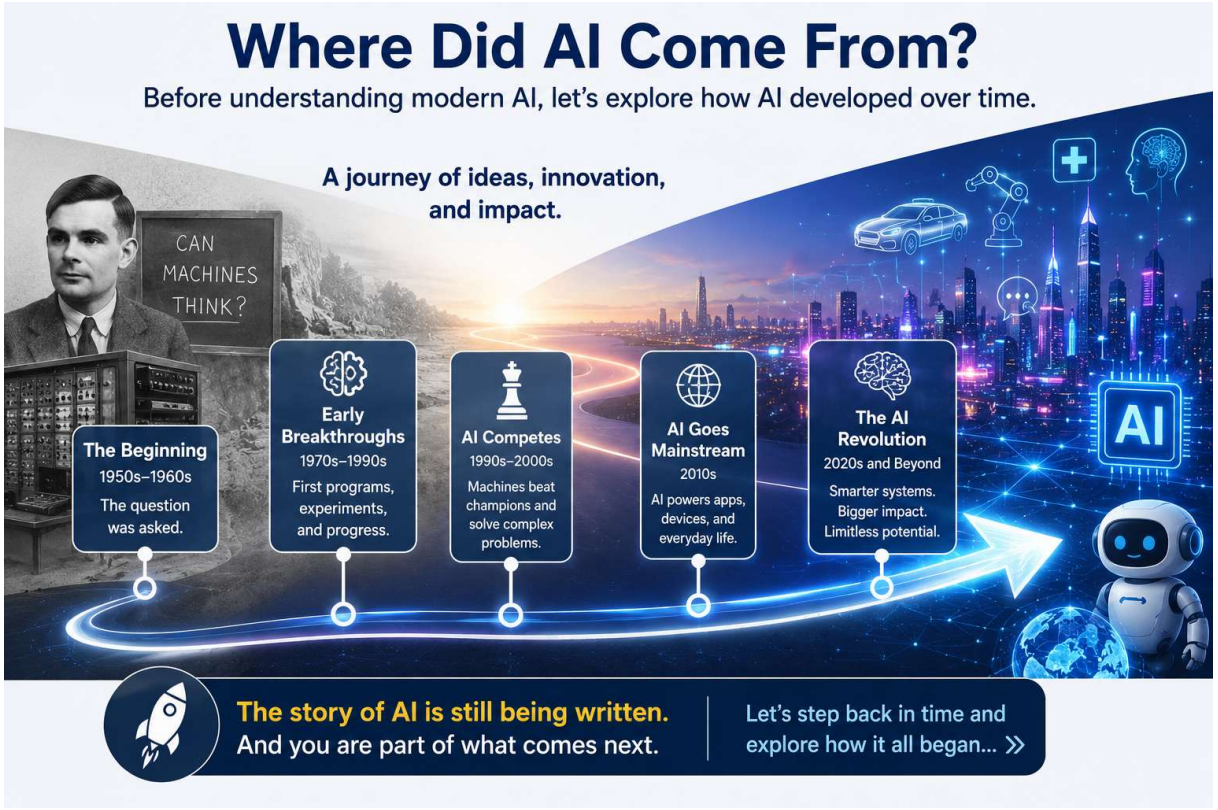
- Bias
- Privacy concerns
- Security risks
- Misinformation
- Job disruption
- Deepfakes
- Misinformation





Where Did AI Come From?

Before understanding modern AI, let's explore how AI developed over time.



Alan Turing

Often called "The Father of Modern Computing"

1950 Published: Computing Machinery and Intelligence

Asked:

- Can Machines

Think?



The Turing Test

- A machine passes the Turing Test if:
- A human cannot reliably distinguish the machine from another human through conversation.

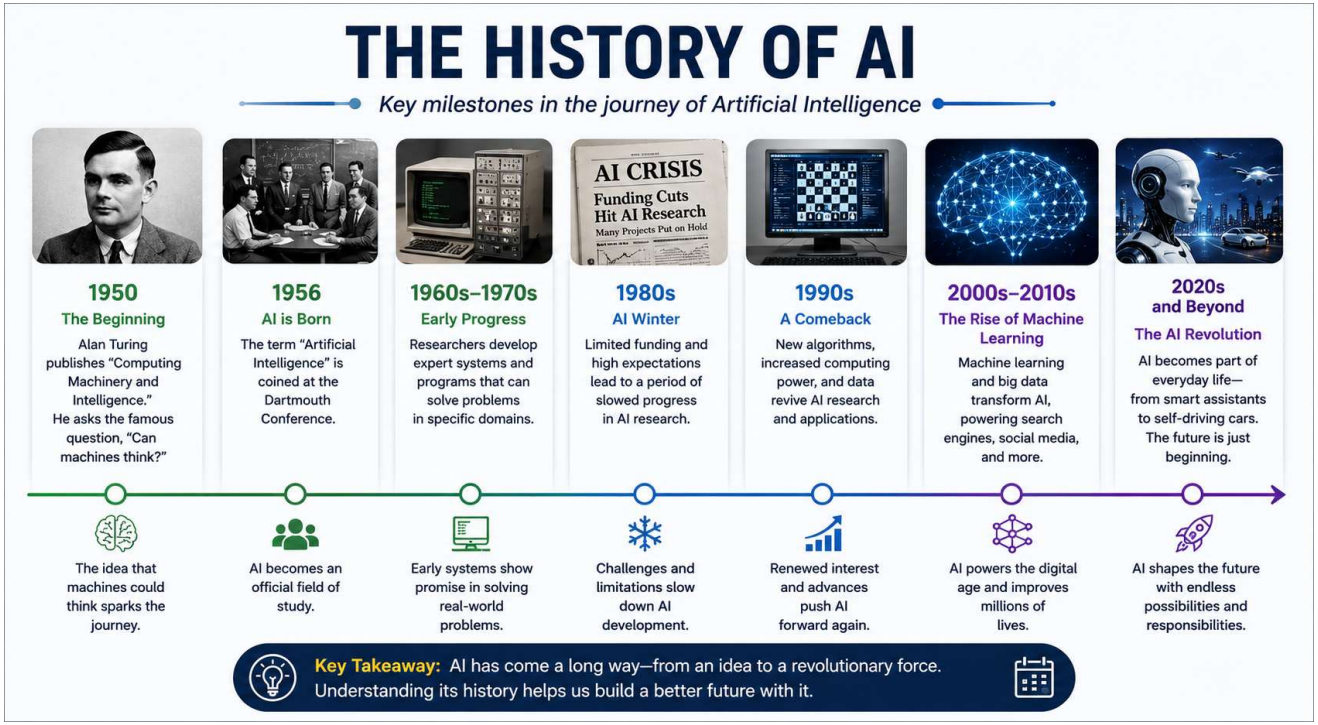
Activity

- Would ChatGPT pass the Turing Test?

Birth of Artificial Intelligence

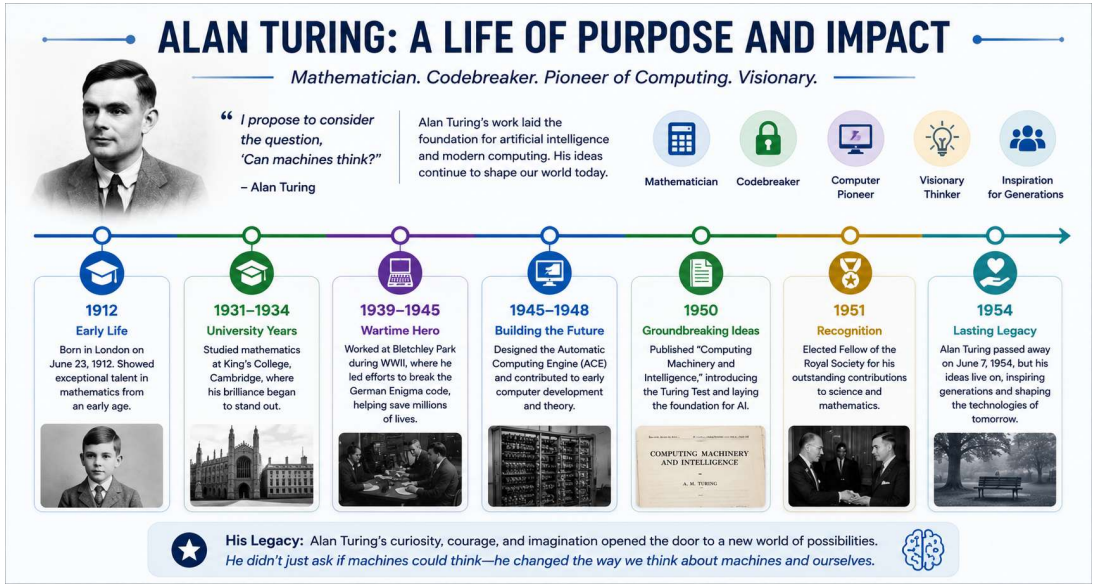
- 1956 -Dartmouth Conference

John McCarthy coined Artificial Intelligence



AI Timeline

- 1950 – Turing
- 1956 – Dartmouth
- 1966 – ELIZA
- 1997 – Deep Blue
- 2011 – Watson
- 2012 – AlexNet
- 2016 – AlphaGo
- 2022+ – ChatGPT Era



ELIZA: The First Famous Chatbot

- Created in 1966
- Simulated a therapist conversation

Why is it Important?

- One of the first conversational AI systems

Some people often assume modern chatbots are new. ELIZA demonstrates conversational AI existed almost 60 years ago.

Deep Blue vs Kasparov

1997: AI defeated world chess champion Garry Kasparov

Why It Mattered: Demonstrated machine problem-solving



IBM Watson

2011: Won Jeopardy

Demonstrated:

- NLP
- Question answering
- Knowledge retrieval



AlphaGo

- 2016: Defeated world champion Lee Sedol

Why It Mattered: Go is vastly more complex than chess



Modern AI Revolution

Powered by:

- Big Data
- GPUs
- Cloud Computing
- Deep Learning

Generative AI Examples

- ChatGPT
- Gemini
- Claude
- DALL-E
- Midjourney

Not All AI Is the Same…

- AI, Machine Learning, and Deep Learning are related but different concepts.

AI vs ML vs DL

Artificial Intelligence

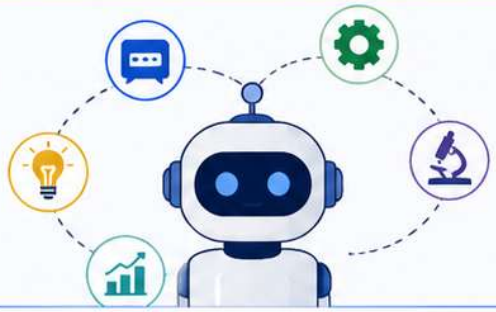
Machine Learning

Deep Learning

Artificial Intelligence (AI)

The broadest field. AI refers to machines that can perform tasks that normally require human intelligence.

Examples: Robotics, expert systems, planning, problem-solving, natural language understanding.



Machine Learning (ML)

A subset of AI. ML enables computers to learn from data and improve their performance without being explicitly programmed.

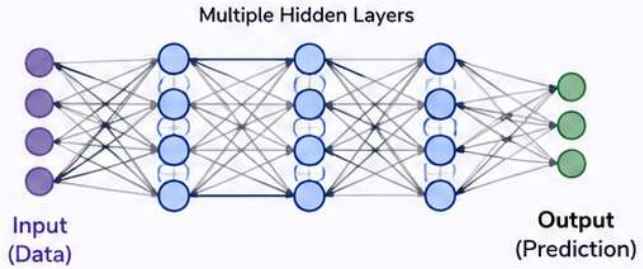
Examples: Recommendation systems, spam detection, fraud detection, predictive analytics.



Deep Learning (DL)

A subset of ML. DL uses artificial neural networks with many layers to learn complex patterns from large amounts of data, especially unstructured data.

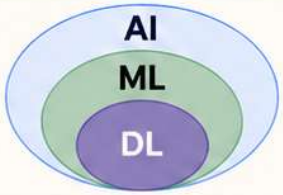
Examples: Image recognition, speech recognition, self-driving cars, ChatGPT.



Key Takeaway

All Deep Learning is Machine Learning.

All Machine Learning is Artificial Intelligence.



AI is the big field.

ML is inside AI.

DL is inside ML.

Artificial Intelligence

Broad field focused on intelligent systems.

Includes:

- Expert systems
- Search algorithms
- Robotics
- Machine Learning

Machine Learning

Machines learn patterns from data.

Examples:

- Spam filtering
- Fraud detection
- Product recommendations

Deep Learning

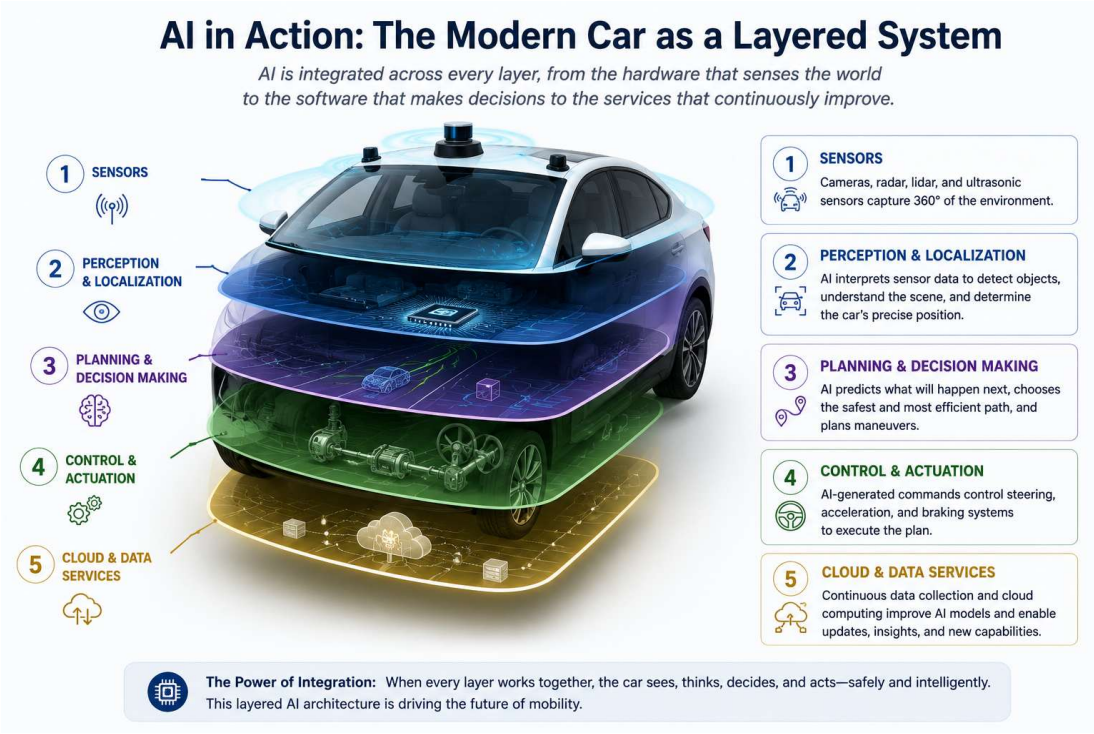
Uses neural networks to identify complex patterns.

Examples:

- Face recognition
- Speech recognition
- Generative AI

Real-World Example: Self-Driving Car

- AI: Entire system
- Machine Learning: Learns driving behavior
- Deep Learning: Recognizes objects



Knowledge Check

Which is broader?

A. Deep Learning

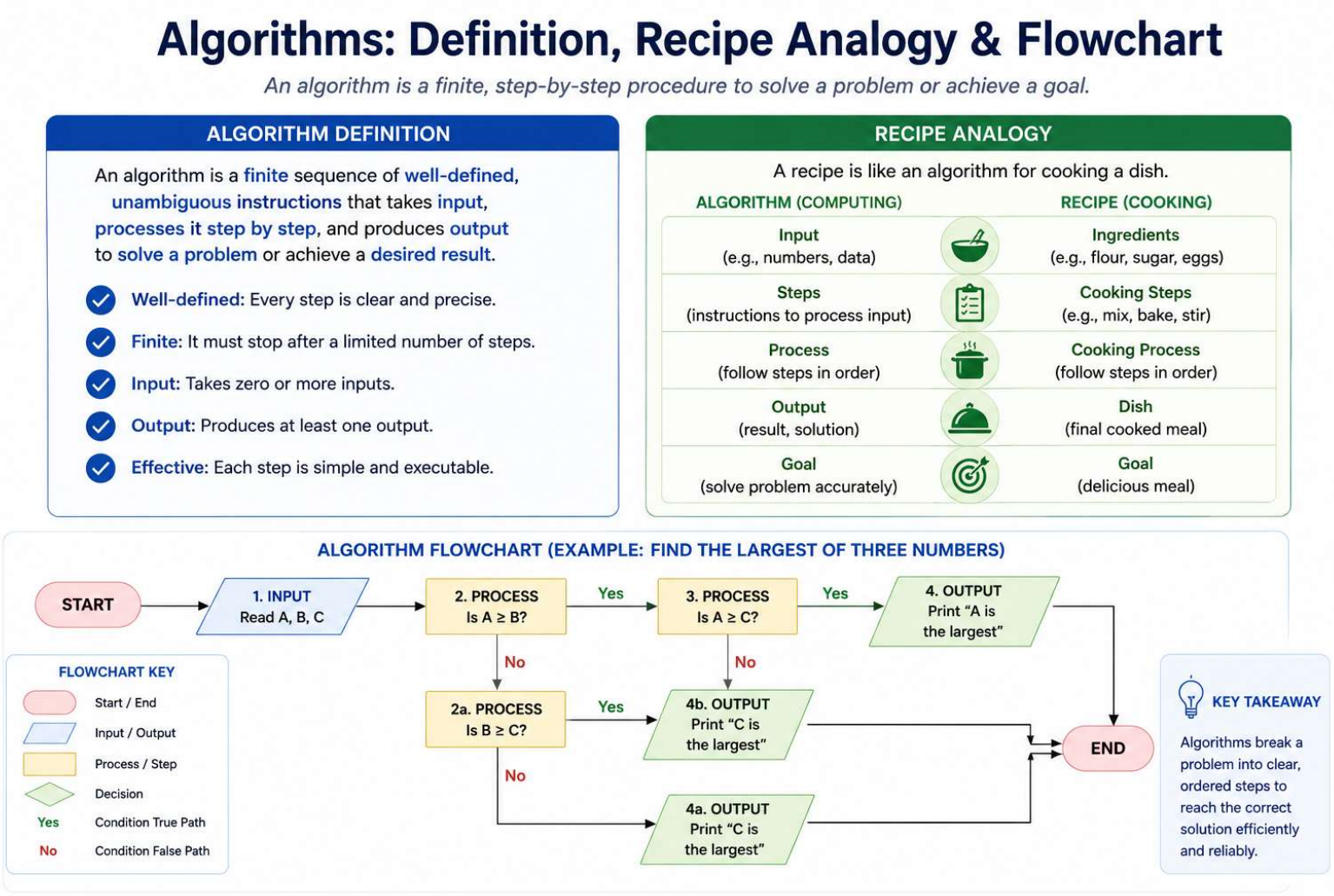
B. Machine Learning

C. Artificial Intelligence

Before We Continue...

Let's learn some important AI vocabulary.

Algorithm


Algorithms: Definition, Recipe Analogy & Flowchart
An algorithm is a finite, step-by-step procedure to solve a problem or achieve a goal.
ALGORITHM DEFINITION
An algorithm is a finite sequence of well-defined, unambiguous instructions that takes input, processes it step by step, and produces output to solve a problem or achieve a desired result.
Well-defined: Every step is clear and precise.
Finite: It must stop after a limited number of steps.
Input: Takes zero or more inputs.
Output: Produces at least one output.
Effective: Each step is simple and executable.
RECIPE ANALOGY
A recipe is like an algorithm for cooking a dish.
ALGORITHM (COMPUTING)
RECIPE (COOKING)
Input (e.g., numbers, data)
Ingredients (e.g., flour, sugar, eggs)
Steps (instructions to process input)
Cooking Steps (e.g., mix, bake, stir)
Process (follow steps in order)
Cooking Process (follow steps in order)
Output (result, solution)
Dish (final cooked meal)
Goal (solve problem accurately)
Goal (delicious meal)
ALGORITHM FLOWCHART (EXAMPLE: FIND THE LARGEST OF THREE NUMBERS)
START
1. INPUT Read A, B, C
2. PROCESS Is A ≥ B?
Yes
3. PROCESS Is A ≥ C?
Yes
4. OUTPUT Print "A is the largest"
No
No
2a. PROCESS Is B ≥ C?
Yes
4b. OUTPUT Print "C is the largest"
No
4a. OUTPUT Print "C is the largest"
END
FLOWCHART KEY
Start / End
Input / Output
Process / Step
Decision
Yes
Condition True Path
No
Condition False Path
KEY TAKEAWAY
Algorithms break a problem into clear, ordered steps to reach the correct solution efficiently and reliably.

Data

DATA: Definition & Examples

DEFINITION: Data is raw, unprocessed facts and figures collected from various sources. It can be numbers, text, images, audio, videos, or any other form of information that can be recorded and stored.

TYPES OF DATA (WITH EXAMPLES)

Numerical Data	Numbers and quantities. Example: 25, 3.14, -7, 1000
Textual Data	Words, sentences, or characters. Example: "Hello", "AI is amazing", "Data Science"
Image Data	Photos, diagrams, charts, etc. Example: JPG, PNG, X-ray images
Audio Data	Sound or speech recordings. Example: MP3, voice notes, podcasts
Video Data	Videos or moving visuals. Example: MP4, surveillance footage, lectures
Sensor / Time-series Data	Data collected over time from sensors. Example: Temperature readings, stock prices, heart rate

REAL-WORLD EXAMPLES

Netflix Viewing History

Netflix collects data every time you watch a show or movie.

- **What it includes:** titles watched, watch time, pause/rewind behavior, ratings
- **Type of data:** Text, numerical, and time-series
- **How it's used:** Recommends movies and shows you are likely to enjoy



E-commerce Purchases

- **What it includes:** products viewed, items bought, price
- **Type of data:** Text, numerical
- **How it's used:** Personalized offers and recommendations

Healthcare Records

- **What it includes:** patient info, symptoms, test results
- **Type of data:** Text, numerical, image
- **How it's used:** Diagnosis, treatment recommendations

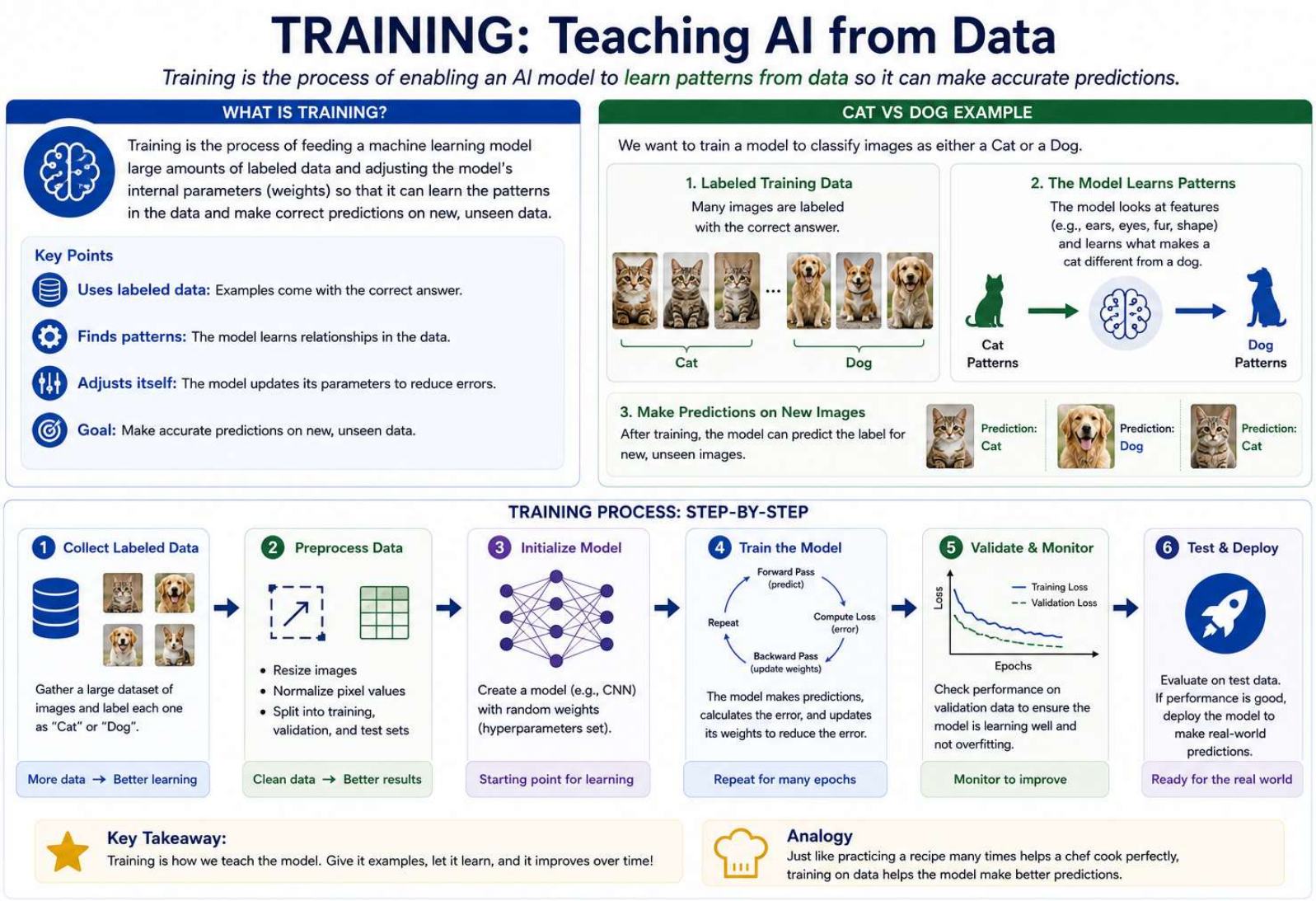
Smartphone Sensors

- **What it includes:** location, steps, battery usage, app usage
- **Type of data:** Numerical, time-series
- **How it's used:** Improve apps and user experience

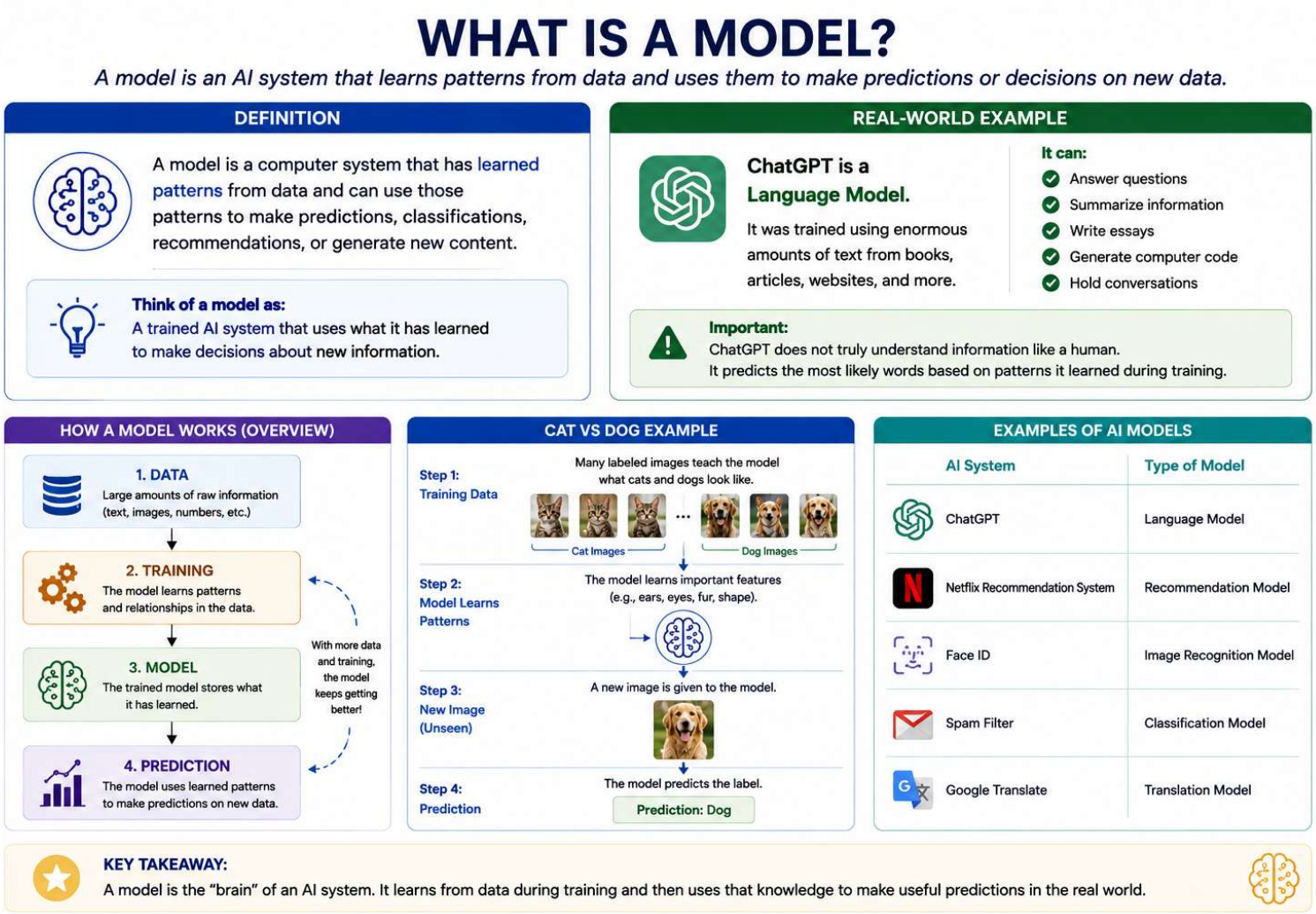
Weather Data

- **What it includes:** temperature, humidity, wind speed, rainfall
- **Type of data:** Numerical, time-series
- **How it's used:** Weather forecasting, alerts

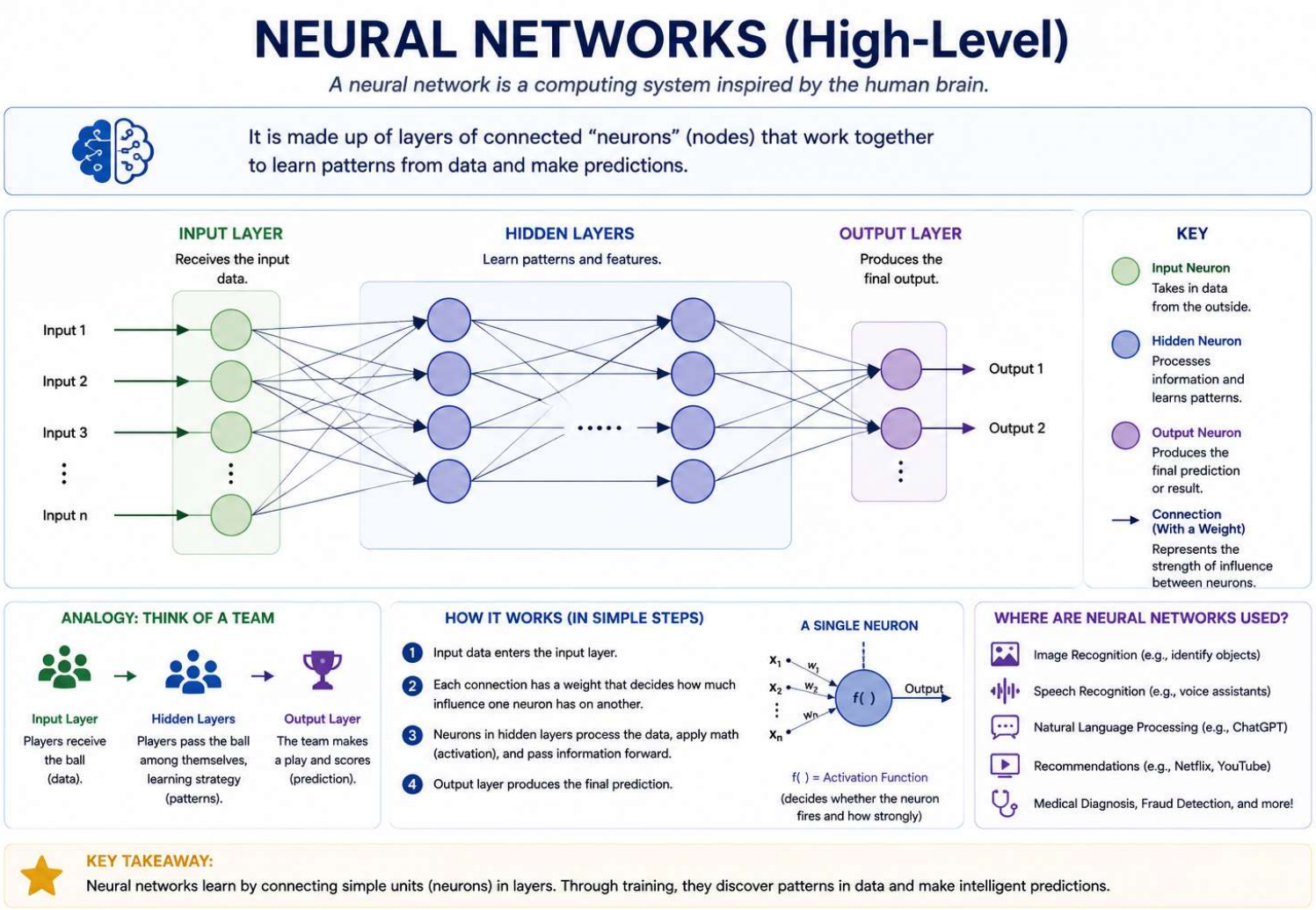
Training


TRAINING: Teaching AI from Data
Training is the process of enabling an AI model to learn patterns from data so it can make accurate predictions.
WHAT IS TRAINING?
Training is the process of feeding a machine learning model large amounts of labeled data and adjusting the model's internal parameters (weights) so that it can learn the patterns in the data and make correct predictions on new, unseen data.
Key Points
Uses labeled data: Examples come with the correct answer.
Finds patterns: The model learns relationships in the data.
Adjusts itself: The model updates its parameters to reduce errors.
Goal: Make accurate predictions on new, unseen data.
CAT VS DOG EXAMPLE
We want to train a model to classify images as either a Cat or a Dog.
1. Labeled Training Data
Many images are labeled with the correct answer.
Cat
Dog
2. The Model Learns Patterns
The model looks at features (e.g., ears, eyes, fur, shape) and learns what makes a cat different from a dog.
Cat Patterns
Dog Patterns
3. Make Predictions on New Images
After training, the model can predict the label for new, unseen images.
Prediction: Cat
Prediction: Dog
Prediction: Cat
TRAINING PROCESS: STEP-BY-STEP
1 Collect Labeled Data
Gather a large dataset of images and label each one as "Cat" or "Dog".
More data → Better learning
2 Preprocess Data
Resize images
Normalize pixel values
Split into training, validation, and test sets
Clean data → Better results
3 Initialize Model
Create a model (e.g., CNN) with random weights (hyperparameters set).
Starting point for learning
4 Train the Model
Forward Pass (predict)
Compute Loss (error)
Backward Pass (update weights)
Repeat
The model makes predictions, calculates the error, and updates its weights to reduce the error.
Repeat for many epochs
5 Validate & Monitor
Loss
Training Loss
Validation Loss
Epochs
Check performance on validation data to ensure the model is learning well and not overfitting.
Monitor to improve
6 Test & Deploy
Evaluate on test data. If performance is good, deploy the model to make real-world predictions.
Ready for the real world
Key Takeaway:
Training is how we teach the model. Give it examples, let it learn, and it improves over time!
Analogy
Just like practicing a recipe many times helps a chef cook perfectly, training on data helps the model make better predictions.

Model


WHAT IS A MODEL?
A model is an AI system that learns patterns from data and uses them to make predictions or decisions on new data.
DEFINITION
A model is a computer system that has learned patterns from data and can use those patterns to make predictions, classifications, recommendations, or generate new content.
Think of a model as:
A trained AI system that uses what it has learned to make decisions about new information.
REAL-WORLD EXAMPLE
ChatGPT is a Language Model.
It was trained using enormous amounts of text from books, articles, websites, and more.
It can:
Answer questions
Summarize information
Write essays
Generate computer code
Hold conversations
Important:
ChatGPT does not truly understand information like a human.
It predicts the most likely words based on patterns it learned during training.
HOW A MODEL WORKS (OVERVIEW)
1. DATA
Large amounts of raw information (text, images, numbers, etc.)
2. TRAINING
The model learns patterns and relationships in the data.
3. MODEL
The trained model stores what it has learned.
4. PREDICTION
The model uses learned patterns to make predictions on new data.
With more data and training, the model keeps getting better!
CAT VS DOG EXAMPLE
Step 1: Training Data
Many labeled images teach the model what cats and dogs look like.
Cat Images
Dog Images
Step 2: Model Learns Patterns
The model learns important features (e.g., ears, eyes, fur, shape).
Step 3: New Image (Unseen)
A new image is given to the model.
Step 4: Prediction
The model predicts the label.
Prediction: Dog
EXAMPLES OF AI MODELS
AI System
Type of Model
ChatGPT
Language Model
Netflix Recommendation System
Recommendation Model
Face ID
Image Recognition Model
Spam Filter
Classification Model
Google Translate
Translation Model
KEY TAKEAWAY:
A model is the "brain" of an AI system. It learns from data during training and then uses that knowledge to make useful predictions in the real world.

Neural Networks

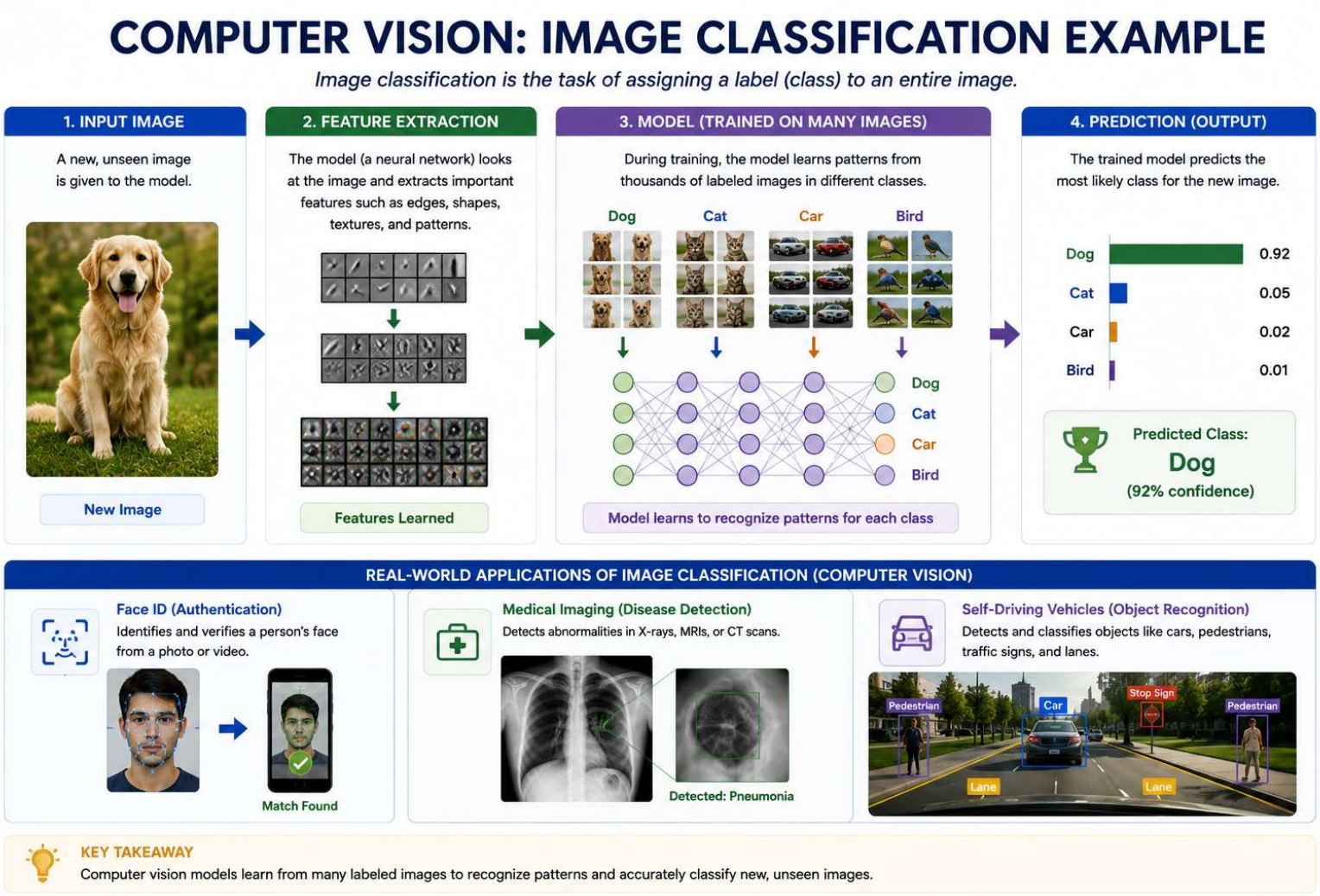


Natural Language Processing

Examples:

- ChatGPT
- Siri
- Google Translate

Computer Vision



What Is an Intelligent Agent?

An intelligent agent is a system that observes what is happening around it, decides what to do, and then acts. An **intelligent agent** is an AI system that can:

- **Perceive** its environment
- **Process** information
- **Make decisions**
- **Take actions** to achieve a goal

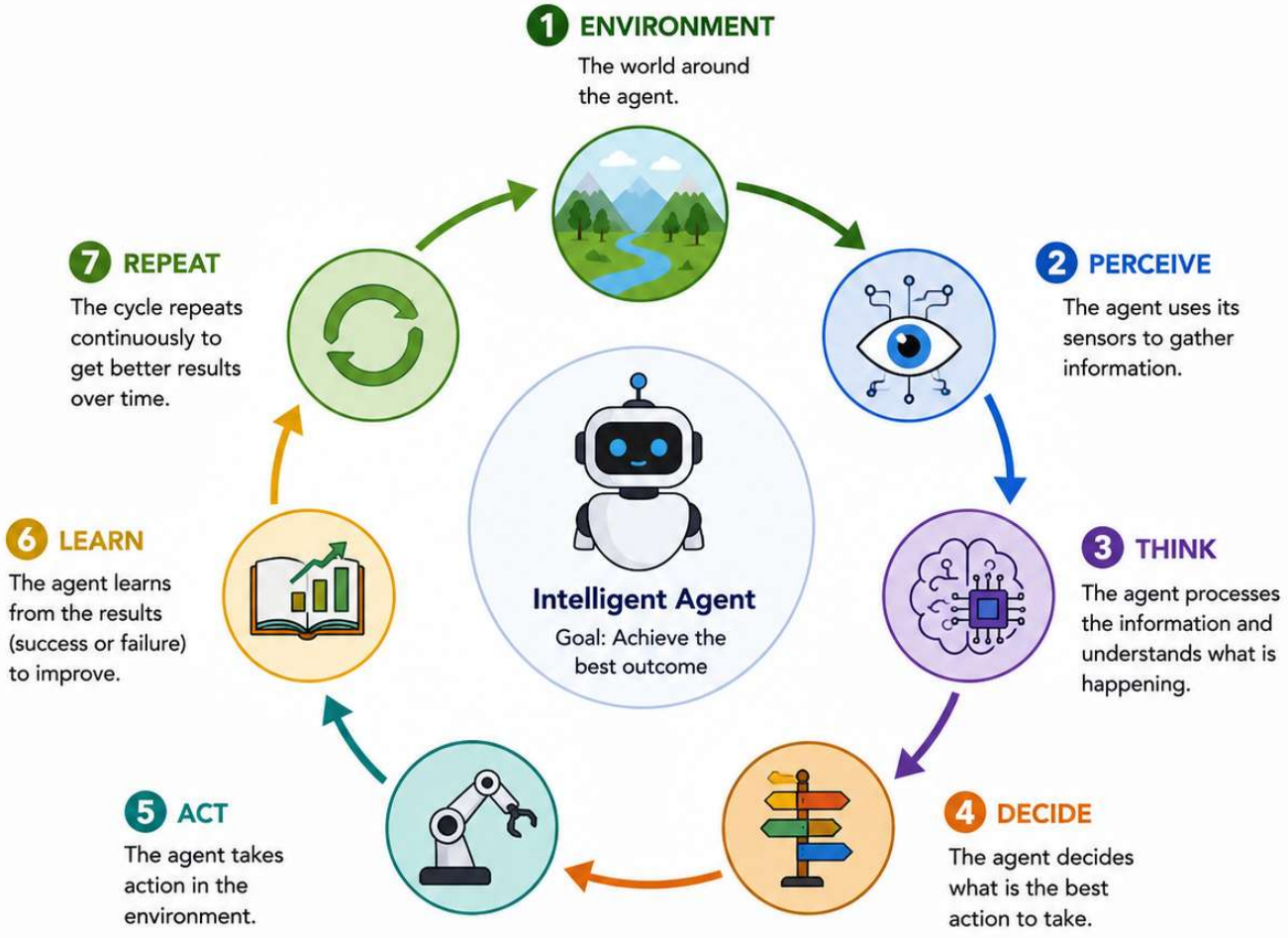
Real-World Examples

- **Roomba:** senses dirt, walls, and furniture, then decides where to clean
- **Alexa:** hears a voice command, processes the request, and gives a response
- **Google Maps:** detects traffic conditions, calculates routes, and recommends the best path

Key Idea: An intelligent agent does not just store information.
It uses information to make decisions and act.

AGENT CYCLE

An intelligent agent continuously interacts with its environment to achieve its goals.



KEY IDEA:
An intelligent agent senses, thinks, acts, and learns in a continuous cycle to achieve its goals in a changing environment.

COMPONENTS OF AN AGENT

An intelligent agent is built from three main components that work together.

1. SENSORS

Sensors perceive the environment and collect information about what is happening.

Examples:
- Camera
- Microphone
- Distance sensor
- Temperature sensor

2. PROCESSOR

The processor (or "brain") interprets the information, makes decisions, and plans action.

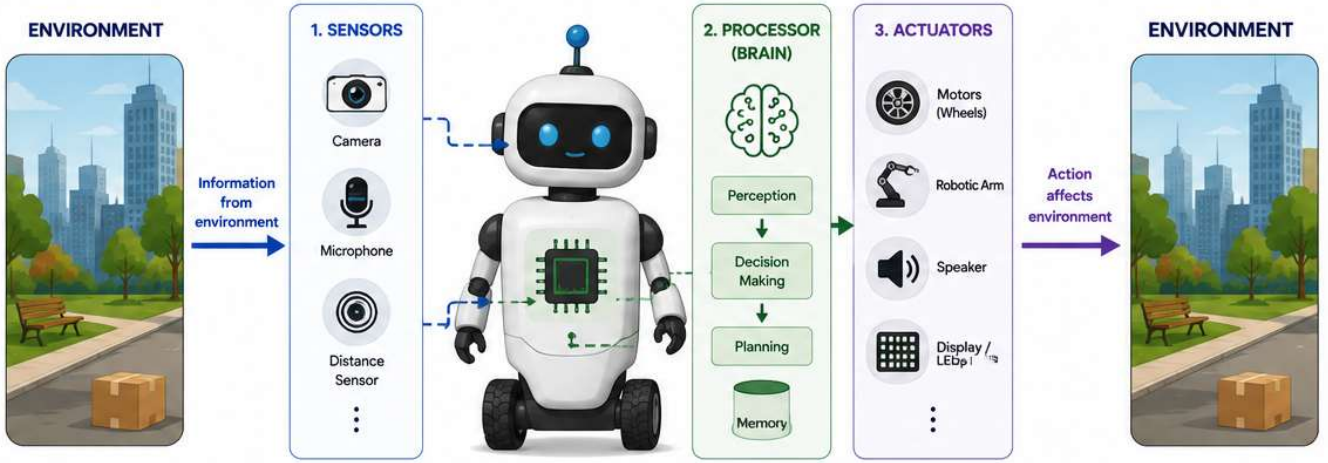
Examples:
- CPU / Microcontroller
- AI model / Algorithm
- Memory

3. ACTUATORS

Actuators carry out the actions chosen by the processor to affect the environment.

Examples:
- Motors
- Wheels
- Robotic arms
- Speakers / Display

ROBOT (AGENT) ARCHITECTURE



Perception: Sensors collect information from the environment. **Cognition:** Processor interprets information, makes decisions, and plans actions. **Action:** Actuators execute the actions in the environment.

This forms a continuous loop: Perceive → Think → Act → Learn → Repeat

REAL-WORLD EXAMPLES



Roomba (Robot Vacuum)
Sensors: Bump sensors, cameras, dirt sensors
Processor: Decides where to clean next
Actuators: Wheels, brushes, vacuum motor



Alexa (Voice Assistant)
Sensors: Microphones
Processor: Understands speech and decides response
Actuators: Speakers, display (echo show)



Google Maps (Navigation Agent)
Sensors: GPS, traffic data, live updates
Processor: Calculates best route
Actuators: Display directions, voice guidance

KEY TAKEAWAY

An agent perceives its environment using sensors, thinks and decides using a processor, and takes actions using actuators to achieve its goals.

Components of an Agent

Component	Purpose
Sensors	Collect information
Processor	Makes decisions
Actuators	Perform actions

Class Activity: Is It an Intelligent Agent?

Directions: For each example in the table in the next slide, determine whether it is an Intelligent Agent.

Ask yourself:

- Does it perceive its environment?
- Does it process information?
- Does it make decisions?
- Does it take actions?

Class Activity: Is It an Intelligent Agent?

System	Perceives Environment?	Makes Decisions?	Takes Actions?	Intelligent Agent?
Roomba Vacuum	Yes	Yes	Yes	Yes
Alexa	Yes	Yes	Yes	Yes
Google Maps	Yes	Yes	Yes	Yes
Calculator	No	No	No	No
Traffic Light Timer	No	Limited	Yes	No
Self-Driving Car	Yes	Yes	Yes	Yes
Thermostat	Yes	Yes	Yes	Yes
Refrigerator	No	No	No	No
Chess AI	Yes	Yes	Yes	Yes
ChatGPT	Yes (through user input)	Yes	Yes	Yes

What Is AGI?

Artificial General Intelligence (AGI) is a hypothetical type of AI that would be capable of understanding, learning, reasoning, and solving problems across many different domains at a level similar to or beyond a human being.

Unlike today's AI systems, AGI would not be limited to a single task. AGI is an AI system that could learn and perform almost any intellectual task that a human can do.

ChatGPT is NOT AGI…

AGI – CURRENT REALITY

Today's AI systems are examples of **Narrow AI (ANI)**.

Examples:

- ChatGPT
- Siri
- Alexa
- Netflix Recommendations
- Google Maps
- Face ID

These systems are very good at specific tasks but cannot do everything a human can do.

What Would AGI Be Able To Do?

An AGI system could potentially:

- Learn new subjects without retraining
- Solve unfamiliar problems
- Transfer knowledge from one field to another
- Understand context deeply
- Adapt to new situations
- Perform many jobs and tasks like humans

Important Fact: AGI does not currently exist. Researchers and companies are actively studying AGI, but today's AI systems are still examples of Narrow AI.

AGI - Real-World Example

Imagine an AI that could:

- Diagnose diseases
- Design bridges
- Write software
- Teach a college course
- Drive a car
- Conduct scientific research

without being specially programmed for each task.
That would be much closer to AGI.

Key Takeaway…

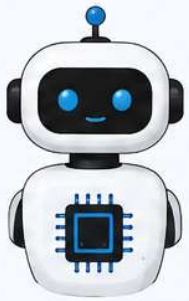
- **Today's AI = Specialized Expert**
- **Future AGI = General Problem Solver**

WHAT IS AGI?

Artificial General Intelligence (**AGI**) is a hypothetical type of AI that can understand, learn, reason, and solve problems across many different domains—at a level similar to or beyond a human being.

NARROW AI (ANI)

Today's AI Systems



- Designed for specific tasks
- Trained for one domain
- Cannot easily transfer skills
- Exists today

Examples:

ChatGPT · Alexa / Siri · Google Maps · Face ID · Netflix Recommendations

HUMAN INTELLIGENCE

Natural Intelligence



- Can perform many different tasks
- Learns continuously from experience
- Transfers knowledge across domains
- Exists today

Examples:



Teacher · Engineer · Scientist · Doctor · Artist

AGI (Artificial General Intelligence)

Future Possibility



- Can perform many different tasks
- Learns continuously and independently
- Transfers knowledge across any domain
- Does not yet exist (research goal)

Future Possibilities:

Diagnose Diseases · Design Bridges · Write Software · Teach Any Subject · Conduct Research · Drive Any Vehicle

TODAY

Narrow AI (ANI)
Specialized AI systems for specific tasks

→

FUTURE RESEARCH

Advancing AI to become more flexible and general

→

FUTURE POSSIBILITY

Artificial General Intelligence (AGI)
General problem solver like a human

KEY TAKEAWAY:
Today's AI is Narrow AI—very good at specific tasks. **AGI** is the long-term goal of creating AI that can do almost anything a human can do.

AI in Movies vs Reality

Movies:

- Terminator
- JARVIS
- Ex Machina
- HAL 9000

Reality:

- ChatGPT
- Siri
- Alexa
- Google Maps

AI in Engineering

AI helps engineers design, build, monitor, and improve complex systems.

Real-World Engineering Examples:

- **Predictive Maintenance:** AI predicts when machines may fail before breakdowns happen.
- **Smart Manufacturing:** AI helps factories adjust production, reduce waste, and improve quality.
- **Robotics:** AI allows robots to sense, move, and complete tasks more independently.
- **Structural Monitoring:** AI analyzes sensor data from bridges, buildings, and roads to detect stress or damage.
- **Design Optimization:** AI helps engineers test many design options quickly before building a prototype.

Key Idea: AI helps engineers move from reacting to problems after they happen to predicting and preventing problems before they happen.

AI IN ENGINEERING

AI helps engineers design, build, operate, and optimize complex systems more efficiently, safely, and sustainably.

MOST IMPORTANT EXAMPLE: PREDICTIVE MAINTENANCE

AI analyzes data from sensors and equipment to predict failures before they happen, reducing downtime, saving money, and improving safety.

1. SMART FACTORY

AI-POWERED MANUFACTURING



- AI optimizes production schedules in real time
- Detects defects automatically using computer vision
- Reduces waste and energy consumption
- Improves quality, efficiency, and safety

2. WIND TURBINE MONITORING

AI FOR RENEWABLE ENERGY



- Sensors collect data on vibration, temperature, wind speed, and more
- AI detects unusual patterns and predicts failures
- Maintenance is scheduled before problems occur
- Improves reliability and reduces repair costs

3. INDUSTRIAL ROBOTS

AI FOR AUTOMATION AND PRECISION



- AI vision systems identify and locate parts
- Robots adapt to changes and work with accuracy
- Improves safety by performing dangerous tasks
- Increases productivity and consistency

KEY TAKEAWAY

AI enables engineers to predict problems, optimize performance, automate tasks, and build smarter, safer, and more sustainable systems.

REAL-WORLD IMPACT

Lower Costs | Less Downtime | Safer Operations | Sustainability

AI in Healthcare

AI supports healthcare professionals by helping them analyze medical data faster and more accurately.

Real-World Healthcare Examples:

- **Cancer Detection:** AI can help identify possible tumors in medical images.
- **Medical Imaging:** AI assists with X-rays, MRIs, CT scans, and pathology images.
- **Drug Discovery:** AI helps researchers search for new medicines more efficiently.
- **Personalized Medicine:** AI can help match treatments to a patient's medical history and genetic information.
- **Patient Monitoring:** AI can detect warning signs from heart rate, oxygen levels, or other health data.

Important Note

AI does not replace doctors!

It supports doctors by helping them notice patterns and make better-informed decisions.

Key Idea: AI can improve healthcare, but human judgment, ethics, and patient safety remain essential.

AI in Environmental Science

AI helps scientists study the environment by analyzing large amounts of data from satellites, sensors, weather stations, and field observations.

Real-World Environmental Examples:

- **Climate Modeling:** AI helps predict climate patterns and extreme weather events.
- **Wildlife Tracking:** AI analyzes camera images, sounds, and GPS data to monitor animal populations.
- **Energy Optimization:** AI helps buildings and power grids use energy more efficiently.
- **Wildfire Prediction:** AI analyzes weather conditions, vegetation, satellite imagery, and historical fire data to identify areas at high risk for wildfires and support early warning systems.
- **Disaster Prediction:** AI supports early warning systems for hurricanes, floods, wildfires, and droughts.
- **Pollution Monitoring:** AI can detect patterns in air quality, water quality, and environmental health data.

Key Idea: AI helps environmental scientists understand complex systems and make faster decisions to protect people, wildlife, and natural resources.

AI Careers

AI is creating career opportunities across STEM fields, not only in computer science.

AI literacy is becoming important even for careers that are not AI-specific. (Engineers, Scientists, Healthcare professionals, Business analysts)

Key Idea: You do not all need to become AI engineer, but almost every STEM career will require some level of AI literacy.

Discussion Question

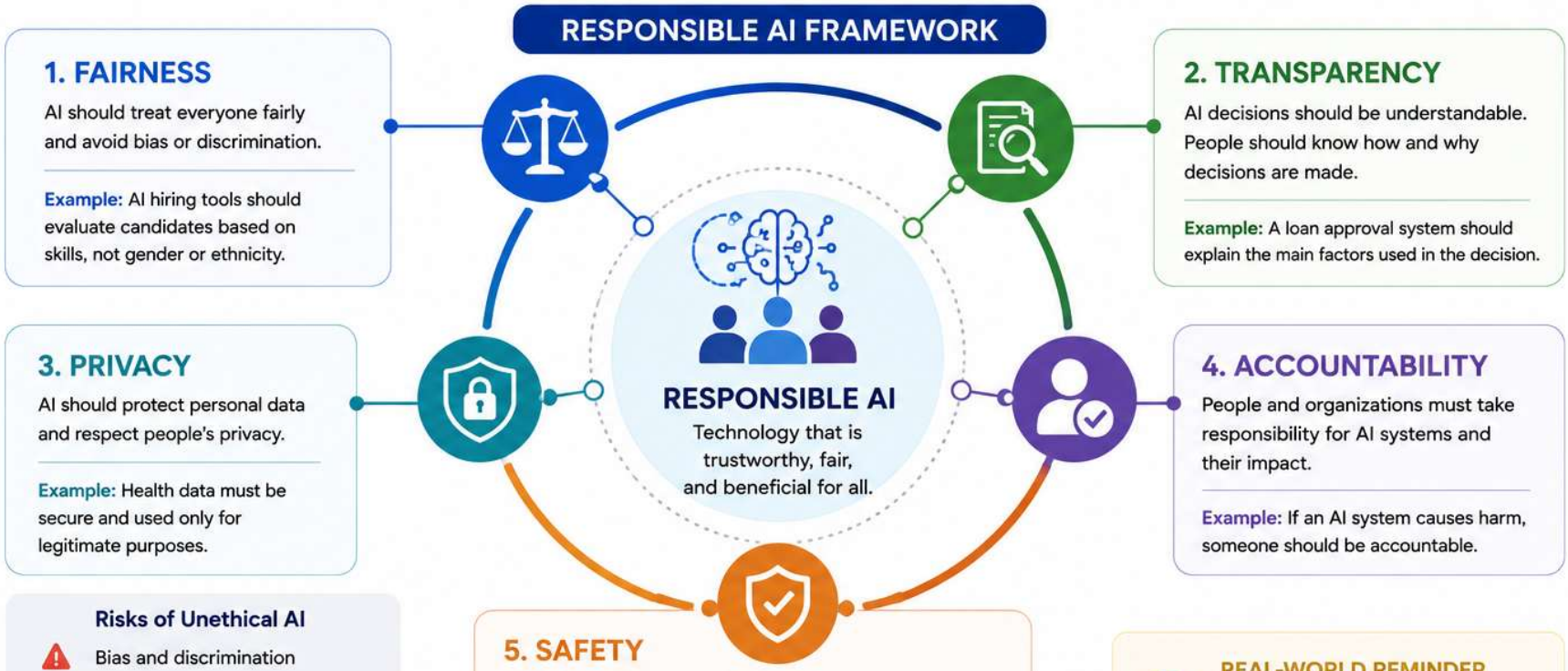
- Which AI-related career connects most closely to your major or career goal? See next slide…

Examples of AI-Related Careers

Career	What They Do
AI Engineer	Builds AI systems and applications
Machine Learning Engineer	Creates models that learn from data
Data Scientist	Analyzes data to find patterns and insights
Robotics Engineer	Designs robots that sense, decide, and act
AI Researcher	Studies new AI methods and technologies
Cybersecurity Analyst	Uses AI to detect threats and protect systems
Health Informatics Specialist	Uses AI and data to improve healthcare
AI Ethics Specialist	Studies fairness, privacy, safety, and accountability

WHY ETHICS MATTERS

AI systems affect people's lives. Ethical AI ensures technology is used responsibly, fairly, and safely for the benefit of everyone.



1. FAIRNESS

AI should treat everyone fairly and avoid bias or discrimination.

Example: AI hiring tools should evaluate candidates based on skills, not gender or ethnicity.

2. TRANSPARENCY

AI decisions should be understandable. People should know how and why decisions are made.

Example: A loan approval system should explain the main factors used in the decision.

3. PRIVACY

AI should protect personal data and respect people's privacy.

Example: Health data must be secure and used only for legitimate purposes.

4. ACCOUNTABILITY

People and organizations must take responsibility for AI systems and their impact.

Example: If an AI system causes harm, someone should be accountable.

Risks of Unethical AI

- Bias and discrimination
- Privacy violations
- Lack of transparency
- No accountability
- Unsafe or harmful outcomes

5. SAFETY

AI systems should be reliable, secure, and designed to avoid harm to users and society.

Example: Self-driving cars must be tested to keep passengers and pedestrians safe.

REAL-WORLD REMINDER

Ethical AI builds trust, protects people, and ensures technology benefits society. Good intentions are not enough—we must design AI responsibly.

KEY TAKEAWAY

Ethics is not optional in AI. Building fair, transparent, private, accountable, and safe AI systems is essential for a better future.

Why Ethics Matters

Which ethical principle do you think is most important?

Why?

Module 1 Summary and Exit Ticket…

What We Learned Today…

By the end of this module, you should be able to:

✓ Define Artificial Intelligence (AI)

✓ Explain major milestones in AI history

✓ Distinguish between AI, Machine Learning, and Deep Learning

✓ Identify common AI vocabulary and concepts

✓ Describe Intelligent Agents and how they interact with their environment

✓ Explain the difference between Narrow AI and Artificial General Intelligence (AGI)

✓ Recognize how AI is used across STEM disciplines

✓ Discuss ethical considerations surrounding AI technologies

Big Ideas to Remember

AI Is Already Part of Daily Life

- From Google Maps and Netflix to Face ID and ChatGPT, AI influences many decisions and experiences every day.

AI Learns From Data

- Data, training, and models are the foundation of modern AI systems.

AI Is Powerful but Not Human

- Current AI systems are specialized tools designed for specific tasks.

Ethics Matters

- As AI becomes more powerful, society must ensure it is fair, transparent, safe, accountable, and respectful of privacy.

Exit Ticket

Before leaving class, answer the following questions:

Question 1

- What is one example of AI that you use regularly?

Question 2

- In your own words, what is the difference between AI, Machine Learning, and Deep Learning?

Question 3

- What is one ethical concern related to AI?

Question 4

- Which AI application or career interested you the most today?

Connecting Today's Learning to Your Assignments

AI Definitions Scavenger Hunt

- Apply the key terminology learned in today's lecture.

AI vs ML vs DL Visual Essay

- Demonstrate your understanding of how these concepts relate to one another.

Everyday AI Discussion

- Analyze how AI affects your daily life and decision-making.

Ethics in AI Discussion

- Evaluate the opportunities and risks associated with responsible AI development.

Looking Ahead…

Today we learned what AI is. In Module 2, we will begin exploring how AI systems think, reason, and make decisions.

Next Module:

- **Knowledge Representation and Reasoning**

Next, we will explore:

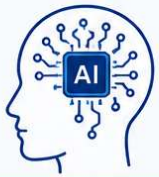
- How machines represent knowledge
- How AI systems make decisions
- Rules, logic, and reasoning
- The foundations of intelligent behavior

Key Takeaway…

AI is transforming science, engineering, healthcare, business, and society.

Understanding AI is no longer optional for STEM professionals—it is becoming an essential skill for the future.

THE BIG PICTURE: WHAT WE LEARNED IN MODULE 1



ARTIFICIAL INTELLIGENCE (AI)

The science and engineering of creating machines that can think, learn, and make decisions.

Key Concepts:
- What is AI?
- History & Milestones
- AI vs ML vs DL
- Key Vocabulary

DATA → TRAINING → MODELS

AI systems learn from data and use models to make predictions or decisions.

DATA
Raw facts, images, text, sensor data

TRAINING
The model learns patterns from the data

MODEL
The trained model makes predictions or decisions

INTELLIGENT AGENTS



AI systems (agents) perceive their environment, think, decide, act, and learn in a continuous cycle.

PERCEIVE
Sensors collect information

THINK
Process information

DECIDE
Choose the best action

ACT
Take action in the environment

LEARN
Improve based on results

APPLICATIONS ACROSS STEM

AI is used in many fields to solve real-world problems and improve lives.

ENGINEERING
- Smart Manufacturing
- Predictive Maintenance
- Robotics

HEALTHCARE
- Cancer Detection
- Medical Imaging
- Drug Discovery

ENVIRONMENT
- Climate Modeling
- Wildlife Tracking
- Energy Optimization

BUSINESS
- Automation
- Customer Insights
- Fraud Detection

EDUCATION
- Personalized Learning
- Intelligent Tutoring
- Student Support

TRANSPORTATION
- Self-Driving Vehicles
- Traffic Optimization
- Route Planning

ETHICS & RESPONSIBILITY

As AI grows more powerful, we must ensure it is developed and used responsibly.

FAIRNESS
Treat everyone equitably and avoid bias

TRANSPARENCY
Be open and clear about how AI makes decisions

PRIVACY
Protect personal data and respect individual rights

ACCOUNTABILITY
People and organizations are responsible for AI systems and outcomes

SAFETY
Design AI to be reliable, secure, and beneficial

FUTURE CAREERS & IMPACT

AI is creating new opportunities and reshaping the future of work.

AI Engineer | ML Engineer | Data Scientist | Robotics Engineer | AI Researcher | AI Ethics Specialist

★ THE BOTTOM LINE

AI is transforming the world. Understanding AI concepts, applications, and ethics prepares you to build a better future.